

Ahmed Mahmoud Abbas

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📅 14/05/2004

📍 Cairo, Egypt

🛡 Military Status: Exempted

🌐 LinkedIn

🐙 GitHub

🔗 Portfolio

Profile

Backend and DevOps Engineer with hands-on production experience in banking infrastructure and security tooling. Built **ACR-QA**, a language-agnostic SAST platform with exploit-verified, ECDSA-attested findings at \$0 recurring cost. Delivered 4 production-handoff prototypes at **NBK Egypt** across AI/ML, banking infrastructure, and secure document delivery. Engineered a Kubernetes observability stack at **225 req/sec, 62ms P95** under Egypt's national DEPI program. Available for remote and on-site backend/DevOps roles.

Professional Experience

DevOps Engineering Trainee

06/2025 – 12/2025 | Zamalek, Giza, Egypt

Digital Egypt Pioneers Initiative (MCIT – EYouth).

- Selected for **MCIT's** national DevOps training program.
- Automated a **5-stage CI/CD pipeline** (Git, Jenkins, GitHub Actions, Docker) and built reusable **Terraform/Ansible** IaC modules provisioning **5+ environments**.
- Deployed, monitored, and optimized containerized microservices on **Kubernetes** with **Prometheus/Grafana** observability.
- Capstone: production URL shortener – **2,715 req/s, 99.93% uptime, 62ms P95**.

Research and Development Intern – Development & Application

08/2025 – 09/2025 | New Cairo, Cairo, Egypt

National Bank of Kuwait - Egypt

- Delivered 4 AI/ML and backend prototypes for NBK's R&D division: a **loan risk predictor** (Streamlit, RandomForest, SMOTE) and an **OFAC sanctions screener** (Python, TF-IDF, fuzzy logic).
- Built a **secure PDF retrieval service** (Spring Boot, **RSA/AES**) integrated with **Temenos T24** via Tomcat.
- Built **Outlook→SFTP automation** (Java, JACOB COM bridge, JSch) to extract and transfer email attachments to Linux servers, replacing a manual workflow.
- Applied SOLID principles, design patterns, and Docker across all **4 prototypes** – extended **2 weeks for performance**, all delivered as **production handoff artifacts** with RUNBOOK docs.

Information Technology Intern

08/2024 | Sheikh Zayed, Giza, Egypt

Emkay Foods

- Supported IT operations across **150+ endpoints**: Active Directory, DHCP, VPN/VLAN configuration, and managed switch/firewall setup (**FortiGate, Sophos**).
- Assisted with ERP user provisioning and access management, improving onboarding and ticket turnaround.

Projects

ACR-QA – Automated Code Review Platform

12/2025 – 05/2026

Graduation Project · Python · RAG · Docker · PostgreSQL ·

GitHub Actions

- Language-agnostic SAST platform – 19 detection engines running in parallel (Python, JS/TS, Go), unified by one normalization schema across **327+ rule mappings**, covering 9/10 OWASP Top 10 categories.
- 96.4% Confirmed Tier precision, 100% CVE recall (8/8)** on RealVuln 2026's 22-repo benchmark. Findings are exploit-verified in an isolated Docker sandbox (13 attack categories) and cryptographically signed for auditability.
- Real distributed system under the hood: **FastAPI** control plane (52 async endpoints, SSE), **Celery + Redis** work queue, dynamic sandbox orchestration.
- RAG-grounded explanations from a 66-rule knowledge base, hallucination-checked via 3× semantic-entropy scoring, at **\$0 recurring cost**.
- 3,247 tests at 88% coverage** across dual CI (GitHub Actions + GitLab), with SARIF export and a policy-engine merge gate.

Redis from Scratch (Rust, TCP, Multithreading, Concurrency, Systems Programming)

03/2025 – 06/2025

- Multithreaded in-memory key-value store** (Rust) supporting Strings, Lists, Sets, Sorted Sets, and TTL.
- 16-command** parser across **4 data structures**, including TTL-based expiry and index-bounded LRANGE/ZRANGE.
- Concurrency-safe, atomic operations via **Arc, Mutex**, and Rust's ownership model.

Containerized URL Shortener – Production Monitoring Stack 10/2025 – 11/2025

DEPI Graduation Project · Node.js · Docker · Kubernetes · Prometheus · Grafana · GitHub Actions · Terraform

- Containerized URL shortener REST API (Node.js, Express, SQLite) with custom Prometheus instrumentation – **19 Grafana panels** across **3 dashboards** (monitoring, analytics, system health).
- **2,715 req/s, 99.93% uptime, 62ms P95** under k6 load testing (**100 concurrent users**).
- Full DevOps lifecycle: **5-stage GitHub Actions CI/CD**, Kubernetes (**HPA**, PodDisruptionBudgets), **Terraform IaC**, multi-channel alerting (Slack, Email, Discord).
- Disaster recovery (**15-min RTO**, Docker volume persistence) + security audit – **B+**, zero critical vulnerabilities.

DecodedSound – AI Cultural Translation Engine 02/2026 – 05/2026

Next.js 15 · TypeScript · Groq LLaMA · Whisper · PostgreSQL · Prisma · Docker

- **3-input pipeline** (typed lyrics, audio, YouTube via yt-dlp) → Groq Whisper transcription → **LLaMA 3.3 70B** translation → **4-panel output** (plain translation, line-by-line annotation, slang dictionary, cultural context) – built for SDK (Esdeekid) music's Cape Flats dialect (Kaaps Afrikaans, Xhosa, prison gang codes), unhandled by existing AI models.
- Living **slang** knowledge base: **502 hand-curated terms**, community contributions, AI candidate discovery, admin moderation, auto-downgrade on **3 flags**.
- **12 REST routes**, 3-tier IP-based **sliding-window rate limiting**, YouTube caching for instant repeat translations.

Education

BSc in Computer Science – Software Engineering10/2022 – 06/2026 | South Sinai, Egypt

King Salman International University

- Activities: IEEE, GDSC, ICPC communities, COP27 Youth Volunteer, Huawei ICT Academy
- Achievement: ICPC ECPC 2024 – **led a 3-member team** to Honorable Mention, 2nd place among KSIU teams

Skills

- **Languages:** Rust, C++, Python, Java, TypeScript, JavaScript
- **Backend:** Node.js, Express, FastAPI, Flask, Spring Boot, REST APIs, PostgreSQL, Redis, MongoDB, SQLite, Prisma, JWT, OAuth
- **DevOps:** Docker, Docker Compose, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, GitLab CI, Prometheus, Grafana, Nginx, CI/CD pipelines
- **ML & AI:** scikit-learn, RAG , Ollama, Groq, LLaMA, Whisper, SMOTE, PCA, RandomForest, Pandas, NumPy, Streamlit
- **Security & AppSec:** SAST, exploit verification, ECDSA/cryptographic attestation, SLSA supply chain security, Sigstore/Cosign, RSA/AES encryption, SFTP, CWE mapping, OWASP Top 10, Semgrep, Bandit, Ruff
- **Infra & Networking:** Linux, AWS, SSH, systemd, TCP/IP, DNS, DHCP, VLANs, VPNs, NAT, FortiGate, Sophos, Active Directory
- **Testing & Observability:** pytest, k6, Apache Bench, SARIF v2.1.0, Grafana, load testing, Docker sandboxing
- **Tools & Practices:** Git, OOP, data structures, concurrency, multithreading, microservices, SOLID principles, clean architecture, CI/CD, RAG pipeline design

Languages

ArabicEnglish

Mother TongueFluent